Linear Interpolator

Design a digital circuit that, given an input signal sampled with a sampling period of ***T***, produces a linearly interpolated version of the input signal as output. Use an interpolation factor ***L = 4*** (for each input sample, the interpolator system must insert other **L – 1** samples).

$$y(nT + uT) = (y_{n+1} - y_n)u + y_n \qquad \text{with } u = \frac{k}{L},\ k \in \{0,1,\ldots,L-1\}$$

The interface of the circuit to be designed is as follows:

You are requested to deal with the various possible error situations, documenting the choices made.

The final project report must contain:

- Introduction (circuit description, possible applications, possible architectures, etc.)
- Description of the architecture designed (block diagram, inputs/outputs, etc.)
- VHDL code (with detailed comments) to be attached to the report.
- Test strategy (Test-plan) and related Testbench for verification; a detailed, though not exhaustive, verification is required, including error situations and borderline cases of functioning
- Interpretation of the results obtained in the automatic synthesis/implementation on a Xilinx FPGA platform in terms of maximum clock frequency (critical path), elements used (slice, LUT, etc.) and estimated power consumption. Comment on any warning messages.
- Conclusions